Aliah University

Odd-Semester (Autumn) Examination - 2022

(For 5th Semester BTech. CSE)

Paper Name: Formal Languages and Automata Theory | **Full Marks:** 80

Paper Code: CSEUGPC13 | **Time:** 3 Hrs

Group – A

1. Fill in the blanks with correct one-word answer – (10X1=10)

a) DPDA is ____________ in power than NPDA.

b) A grammar derive ________ a language.

c) A language is a Set ____________ of all possible strings over some alphabet.

d) Type 1 grammars are Context ___________.

e) Stack in PDA has infinite ________ storing capability.

f) Complement of a DCFL is always a DCFL.________(True/False)

g) Moore machine has ____ number of final states.

h) Turing Machines for recursive enumerable languages never guarantee a halt ______ for the strings not of the language.

i) Only the finite languages are regular. ___________(True/False)

j) Minimized DFA of ***L: (aa+aaa)* where*** $\sum=\{a\}$ has n ________ number of states.

Group – B

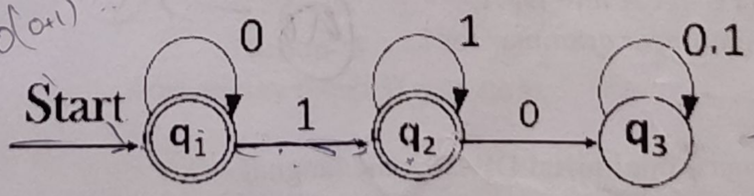
(Answer any 6questions) (6X5=30)

2. What is derivation? Explain RMD and LMD. When a grammar is called ambiguous grammar? Give an example and explain.

3. Using Pumping Lemma prove that –

   L: $\{a^p \mid p$ *is any prime number,* $\sum=\{a\}\}$ is not a Regular Language.

4. Write Arden's theorem. Using Arden's theorem find Regular Expression for the Finite Automata given below.



5. Construct a minimized Deterministic Finite Automata for the language given below -

   L: All strings, not having '101' substring, over alphabet $\sum = \{0, 1\}$

6. Prove that – "Context free languages are closed under concatenation".

7. Construct a Turing Machine for the language below –

   L: $\{a^n b^m \mid n=m, n,m>=1$ *where* $\sum=\{a,b\}\}$

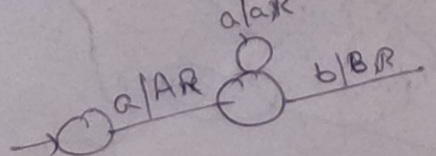
8. Simplify the following Context Free Grammar – here S is starting symbol, terminals are {b,d,p, Ɛ}

   S→ABCd

   A→BC

   B→bB|Ɛ

   C→pC|Ɛ



Group - C

(10X4=40)

(Answer any 4 questions)

9. Consider the language below to answer the following questions –

$$L: \{a^n b^{n+m} c^m \mid n,m >= 1\}$$

[5+5=10]

a) Construct a Push down Automata (PDA) which accepts L.

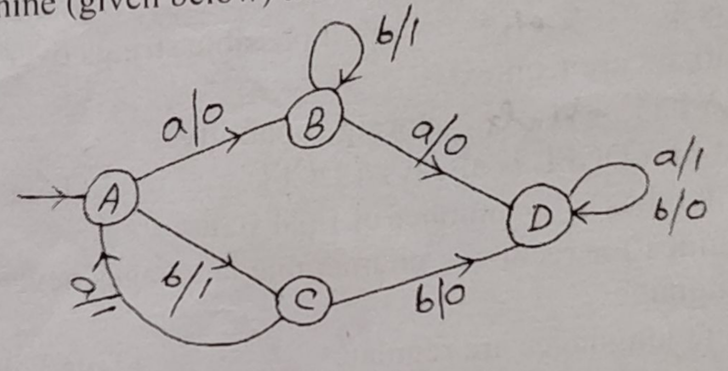
b) Write a Context Free Grammar (CFG) which generates L.

[5+5=10]

10.

a) Write a note on Moore machine and Mealy Machine and highlight the basic differences.

b) Convert the Mealy Machine (given below) into its corresponding Moore Machine.



11. a) Construct a Turing Machine to accept the language given below -

L: {ww | where 'w' means strings formed over inputs ={a,b}}

[8+2=10]

b) Is it a deterministic or non-deterministic Turing machine?

[2.5*4=10]

12. Write short notes on followings-

a) CNF (Chomsky's Normal Form)
b) GNF (Greibach Normal Form)
c) Removal of Left recursion
d) Context Sensitive Grammar (CSG)

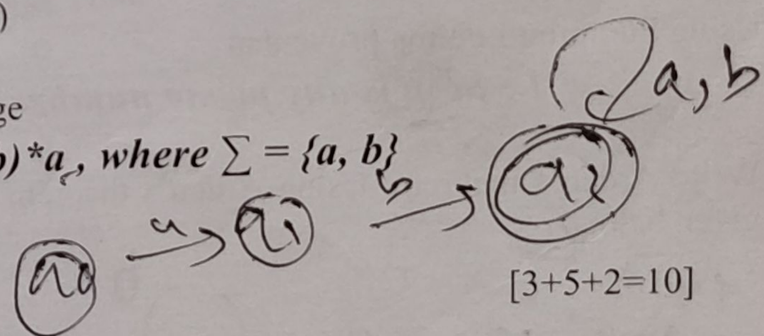
13. A regular expression is given for a language

L: ab(a+b)*a, where Σ = {a, b}

a) Construct $\mathcal{E}$ -NFA for L.

b) Convert the $\mathcal{E}$ -NFA into DFA.

c) Write a Right linear grammar for L.



[3+5+2=10]

14.

a) Construct a minimized DFA for the language -

[5+5=10]

L: All strings, where last two symbols are same, over alphabet Σ = {0, 1}

b) Write Context Free Grammar for the language –

L: {wcw^R | where 'w' means strings formed over inputs ={a,b}, c is a terminal and w^R is the reverse of w}